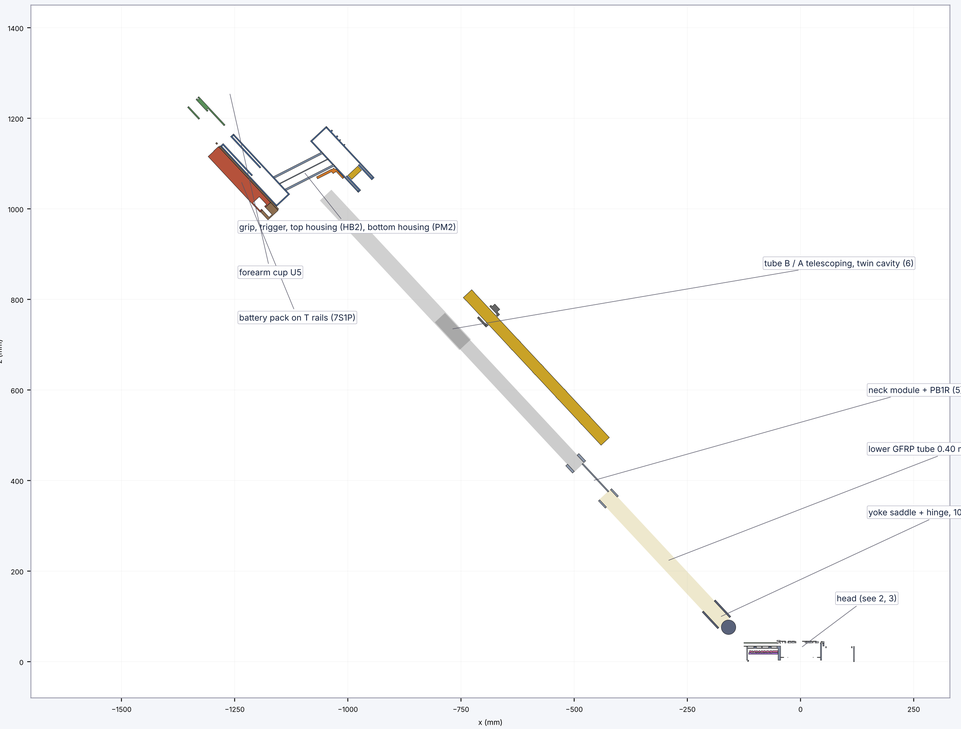
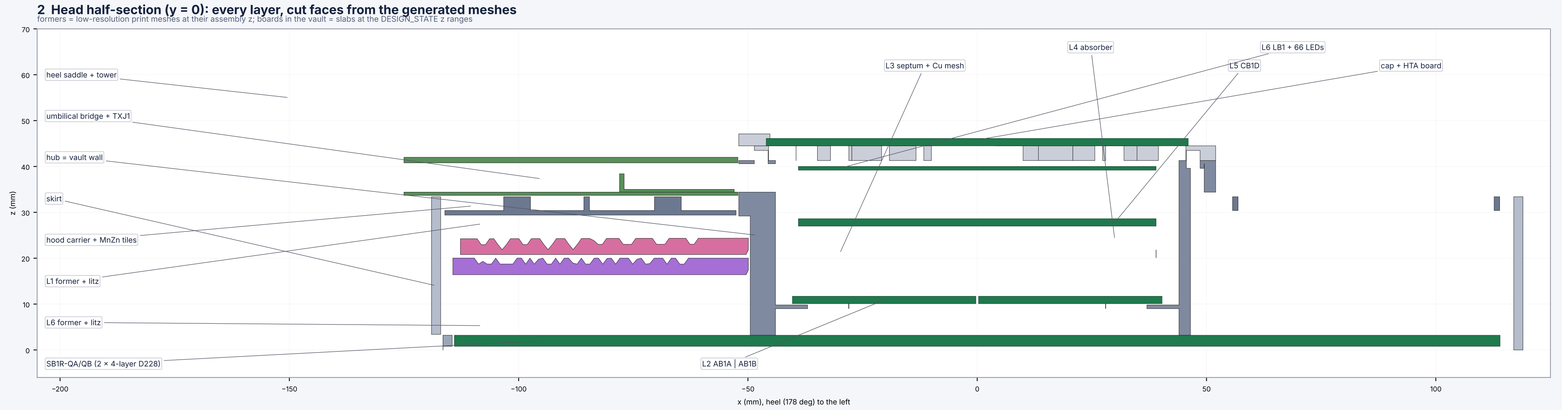
PHI-FOLD 233 - full structure and layered cut-away blueprint (r3): current meshes, boards and parts

1 Whole detector, extended, shaft at 47 deg: mid-plane cut of the current head, yoke, tubes, neck module and upper assembly v1

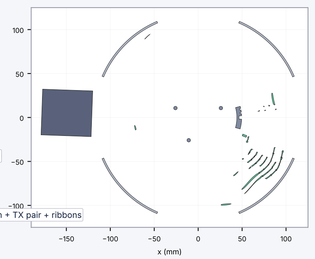


2 Head half-section (y = 0): every layer, cut faces from the generated meshes

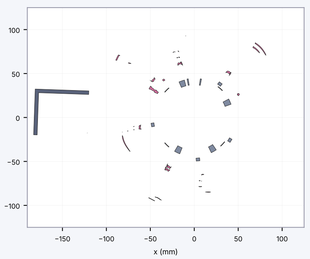
formers = low-resolution print meshes at their assembly z; boards in the vault = slabs at the DESIGN_STATE z ranges



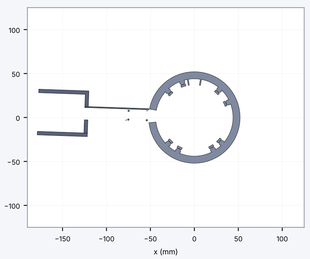
3a Plan section z = 5.2: L6 former (litz petals)



3b Plan section z = 24: absorber cell, hub ribs

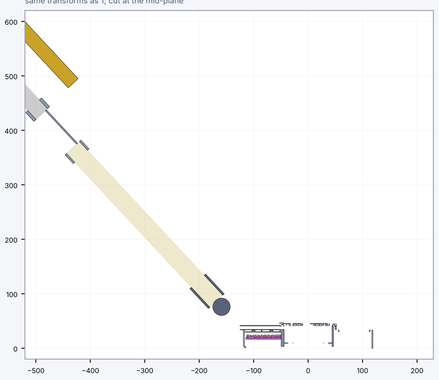


3c Plan section z = 36: CB1D slot, TX-lead channel, exit window



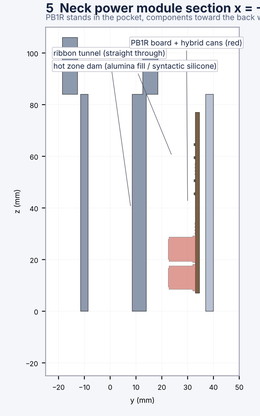
4 Yoke saddle, hinge, lower tube and neck module at 47 deg (zoom of 1)

same transforms as 1; cut at the mid-plane



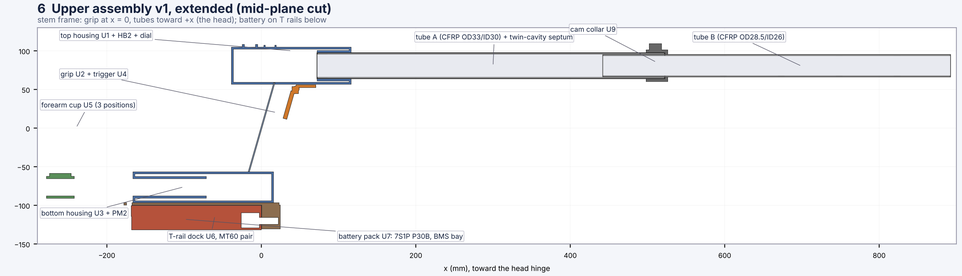
5 Neck power module section x = -7: PB1R with the 9 hybrid cans

PB1R stands in the pocket, components toward the back wall (mech_v20.py); the cans are 10.3 mm tall: pocket d



6 Upper assembly v1, extended (mid-plane cut)

stem frame: grip at x = 0, tubes toward +x (the head); battery on T rails below

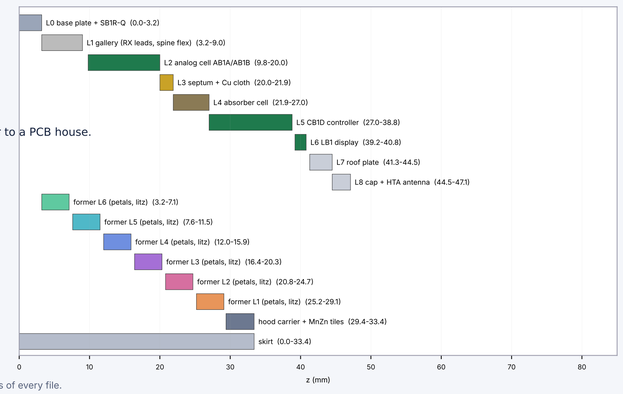


How to read this sheet

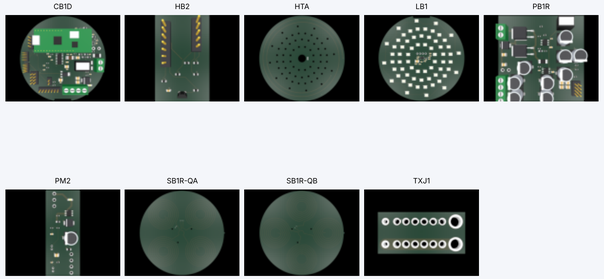
- Panels 1-6 are exact mid-plane / plan cuts of the CURRENT generated meshes (stl/, work/kit_v15/LOW formers, the generated PB1R model).
- Colours: formers L1-L6 orange to green, hub / hood grey-blue, boards green, GFRP tube gold, CFRP tubes black, housing blue, trigger orange, pack red, dock brown.
- Boards inside the vault are slabs at the DESIGN_STATE z ranges (the generators place no board STL there); SB1R-Q sits in the base-plate recess.
- Formers are the LOW-resolution meshes (print meshes: make formers-print, about 70 MB each).
- 47 deg is the use pose; the shaft tubes, neck module and upper assembly use the transforms of render_system.py.
- Panel 8 is KiCad's own 3-D render of the routed board files (copper, mask, pads; no component 3-D models exist).

Status: MODELED design drawing, not a fabrication drawing. See 00_INDEX.md for what is verified, what is open and what must happen before anything is sent to print or to a PCB house.

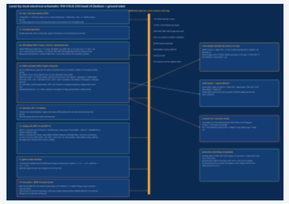
7 Head layer map (z, from the meshes and DESIGN_STATE section 3)



8 Boards (KiCad 3-D render of the routed files, top side)



9 Level-by-level electrical schematic (LEVELS_S



MODELED design drawing: sections computed from the generated STLs; board slabs at the DESIGN_STATE z ranges; nothing here is built or measured. Not a fabrication drawing: see 00_INDEX.md for the status of every file.